

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF APPLIED MATHEMATICS
AMA3640 STATISTICAL INFERENCE

Assignment 1

Due at 11:59pm on 12 October 2025

1. Let $X_1, \dots, X_n$ be a random sample from the density

$$f(x; \theta) = \theta(\theta + 1)x(1 - x)^{\theta-1}, \quad 0 < x < 1,$$

where $\theta > 0$.

- (a) Find the method of moments estimator of θ .
- (b) Find the maximum likelihood estimator (MLE) of θ .
- (c) Find the Cramér–Rao lower bound of the variance of unbiased estimators of $1/\theta$.

2. Let $X_1, \dots, X_n$ be a random sample from the Poisson distribution with parameter $\lambda > 0$:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

- (a) Find the MLE of λ .
- (b) Suppose it is known that $\lambda \geq 1$. Find the MLE of λ under this restriction.

3. Let $X_1, \dots, X_n$ be a random sample from the density

$$f(x; \theta) = 2\theta^2 x^{-3}, \quad \theta \leq x < \infty,$$

where $\theta > 0$.

- (a) Find a one-dimensional sufficient statistic for θ .
- (b) Find the method of moments estimator of θ .
- (c) Find the MLE of θ . Find the bias of the MLE, and show that the MLE is a consistent estimator.

4. Consider a random sample $X_1, \dots, X_n$ with density

$$f(x; \theta) = \theta x^{\theta-1}, \quad 0 < x < 1,$$

where $\theta > 0$.

- (a) Find a one-dimensional sufficient statistic for θ .
- (b) Find the MLE of $1/\theta$, and find its bias.

- (c) Find the Cramér–Rao lower bound for the variance of unbiased estimators of $1/\theta$.
- (d) Verify whether the MLE of $1/\theta$ is the best unbiased estimator for $1/\theta$.

5. Let $X_1, \dots, X_n$ be a random sample from the distribution with density

$$f(x; \theta) = e^{-(x-\theta)}, \quad x > \theta,$$

where $\theta \in \mathbb{R}$.

Consider the following three estimators for θ : $\hat{\theta}_1 = X_1 - 1$, $\hat{\theta}_2 = \bar{X} - 1$, where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, and $\hat{\theta}_3 = \min\{X_1, \dots, X_n\} - \frac{1}{n}$.

- (a) Show that all three estimators are unbiased for θ .
- (b) Evaluate the variances of these estimators.

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Assignment 1 Solution

1. (a) The expectation of X_1 is

$$\begin{aligned} EX_1 &= \int_0^1 \theta(\theta+1)x^2(1-x)^{\theta-1} dx = -(\theta+1) \int_0^1 x^2 d(1-x)^\theta \\ &= (\theta+1) \int_0^1 2x(1-x)^\theta dx - (\theta+1)x^2(1-x)^\theta \Big|_0^1 \\ &= - \int_0^1 2x d(1-x)^{\theta+1} \\ &= 2 \int_0^1 (1-x)^{\theta+1} dx - 2x(1-x)^{\theta+1} \Big|_0^1 \\ &= -\frac{2}{\theta+2}(1-x)^{\theta+2} \Big|_0^1 = \frac{2}{\theta+2}. \end{aligned}$$

The method of moments estimator, $\tilde{\theta}$, solves $E_{\tilde{\theta}}X = \bar{X}$, so

$$\tilde{\theta} = \frac{2(1-\bar{X})}{\bar{X}}.$$

- (b) The log-likelihood function is

$$\log L(\theta) = \sum_{i=1}^n \log f(X_i; \theta) = n \log\{\theta(\theta+1)\} + (\theta-1) \sum_{i=1}^n \log(1-X_i) + \sum_{i=1}^n \log X_i.$$

The first derivative of $\log L$ is

$$\frac{\partial}{\partial \theta} \log L(\theta) = n \frac{2\theta+1}{\theta(\theta+1)} + \sum_{i=1}^n \log(1-X_i).$$

Solving the equation $\frac{\partial}{\partial \theta} \log L(\theta) = 0$ with $\theta > 0$, we have

$$\hat{\theta} = -\frac{\sqrt{4+c^2} + (2+c)}{2c},$$

where $c = n^{-1} \sum_{i=1}^n \log(1-X_i)$. Since

$$\frac{\partial^2}{\partial \theta^2} \log L(\theta) = n \frac{2\theta(\theta+1) - (2\theta+1)^2}{\theta^2(\theta+1)^2} = -n \frac{\theta^2 + (\theta+1)^2}{\theta^2(\theta+1)^2} < 0,$$

$\hat{\theta}$ is the MLE of θ .

- (c) Note that

$$\begin{aligned} \frac{\partial}{\partial \theta} \log f(x; \theta) &= \frac{\partial}{\partial \theta} \{\log \theta + \log(1+\theta) + (\theta-1) \log(1-x) + \log x\} \\ &= \frac{1}{\theta} + \frac{1}{1+\theta} + \log(1-x). \end{aligned}$$

The Fisher information is

$$I(\theta) = -E_{\theta} \left\{ \frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right\} = -E_{\theta} \left\{ -\frac{1}{\theta^2} - \frac{1}{(1+\theta)^2} \right\} = \frac{\theta^2 + (1+\theta)^2}{\theta^2(1+\theta)^2}.$$

The Cramér–Rao lower bound of the variance of unbiased estimators of $1/\theta$ is

$$\frac{1/\theta^4}{nI(\theta)} = \frac{(1+\theta)^2}{n\theta^2(2\theta^2 + 2\theta + 1)}.$$

2. (a) The likelihood function for the Poisson sample is

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{X_i} e^{-\lambda}}{X_i!} = \lambda^{\sum_{i=1}^n X_i} e^{-n\lambda} \prod_{i=1}^n \frac{1}{X_i!}.$$

The log-likelihood function is

$$\log L(\lambda) = \left(\sum_{i=1}^n X_i \right) \log \lambda - n\lambda - \sum_{i=1}^n \log(X_i!).$$

The first derivative is

$$\frac{\partial}{\partial \lambda} \log L(\lambda) = \frac{\sum_{i=1}^n X_i}{\lambda} - n.$$

Setting this equal to zero and solving for λ give

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}.$$

To confirm that this is a maximizer, we compute the second derivative:

$$\frac{\partial^2}{\partial \lambda^2} \log L(\lambda) = -\frac{\sum_{i=1}^n X_i}{\lambda^2} < 0.$$

Since the second derivative is negative, the solution is indeed a maximizer.

- (b) If it is known that $\lambda \geq 1$, the MLE is the maximizer of the likelihood over $\lambda \geq 1$. If $\bar{X} \geq 1$, then $\hat{\lambda} = \bar{X}$ as before. If $\bar{X} < 1$, the unconstrained MLE would be less than 1, but the constraint forces $\lambda \geq 1$. Since the likelihood is decreasing for $\lambda > \bar{X}$, the maximum occurs at $\lambda = 1$. Therefore, the restricted MLE is

$$\hat{\lambda}_{\text{restricted}} = \max\{1, \bar{X}\}.$$

3. (a) We write the density function as $f(x; \theta) = 2\theta^2 x^{-3} I(x \geq \theta)$. The joint density function is

$$\prod_{i=1}^n f(x_i; \theta) = 2^n \theta^{2n} \prod_{i=1}^n x_i^{-3} I(x_i \geq \theta) = 2^n \theta^{2n} \left(\prod_{i=1}^n x_i^{-3} \right) I(\min_i x_i \geq \theta).$$

By the factorization theorem, $\min_i X_i$ is a sufficient statistic for θ .

(b) The expectation of X_1 is

$$EX_1 = \int_{\theta}^{\infty} x \cdot 2\theta^2 x^{-3} dx = 2\theta^2 \int_{\theta}^{\infty} x^{-2} dx = 2\theta.$$

The method of moments estimator is $\hat{\theta}_{\text{MME}} = \bar{X}/2$.

(c) The likelihood function

$$L(\theta) = 2^n \theta^{2n} \left(\prod_{i=1}^n X_i^{-3} \right) I(\theta \leq \min_i X_i)$$

is increasing in θ on $(0, \min_i X_i]$ and is zero on $(\min_i X_i, \infty)$. Therefore, the MLE of θ is $\hat{\theta} = \min_i X_i$. To find the density function of $\min_i X_i$, note that

$$\begin{aligned} F_{(1)}(x) &\equiv P(\min_i X_i \leq x) = 1 - P(\min_i X_i > x) \\ &= 1 - P(X_1 > x, \dots, X_n > x) \\ &= 1 - \left(\int_x^{\infty} 2\theta^2 t^{-3} dt \right)^n = 1 - \left(\frac{\theta}{x} \right)^{2n} \end{aligned}$$

for $x > \theta$. The density function of $\min_i X_i$ is

$$f_{(1)}(x) = \frac{d}{dx} F_{(1)}(x) = 2n\theta^{2n} x^{-(2n+1)} \quad \text{for } x > \theta.$$

We have

$$E(\min_i X_i) = \int_{\theta}^{\infty} 2n\theta^{2n} x^{-(2n+1)} x dx = \frac{2n\theta}{2n-1}.$$

The bias of the MLE is $\theta/(2n-1)$. For any $\epsilon > 0$,

$$P(|\min_i X_i - \theta| > \epsilon) = P(\min_i X_i > \theta + \epsilon) = 1 - F_{(1)}(\theta + \epsilon) = \left(\frac{\theta}{\theta + \epsilon} \right)^{2n}.$$

Since the right-hand side tends to 0 as $n \rightarrow \infty$, the MLE is consistent.

4. (a) The joint density function is

$$\prod_{i=1}^n f(x_i; \theta) = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1}.$$

By the factorization theorem, $\prod_{i=1}^n X_i$ is a sufficient statistic for θ .

(b) The log-likelihood function is

$$\log L(\theta) = \sum_{i=1}^n \log f(X_i; \theta) = n \log \theta + (\theta - 1) \sum_{i=1}^n \log X_i.$$

The first derivative of $\log L$ is

$$\frac{\partial}{\partial \theta} \log L(\theta) = \frac{n}{\theta} + \sum_{i=1}^n \log X_i.$$

Solving the equation $\frac{\partial}{\partial \theta} \log L(\theta) = 0$ with $\theta > 0$, we have

$$\hat{\theta} = -\frac{n}{\sum_{i=1}^n \log X_i}.$$

Since

$$\frac{\partial^2}{\partial \theta^2} \log L(\theta) = -\frac{n}{\theta^2} < 0,$$

$\hat{\theta}$ is the MLE of θ . Thus, by the invariance property of the MLE, the MLE of $\frac{1}{\theta}$ is

$$\frac{1}{\hat{\theta}} = -\frac{1}{n} \sum_{i=1}^n \log X_i.$$

Note that

$$E(\log X_1) = \int_0^1 \theta x^{\theta-1} \log x \, dx = \int_0^1 \log x \, dx^\theta = -\int_0^1 x^\theta \, d \log x = -\int_0^1 x^{\theta-1} \, dx = -\frac{1}{\theta}.$$

We have $E\left(-\frac{1}{n} \sum_{i=1}^n \log X_i\right) = 1/\theta$. The bias is 0.

(c) The Fisher information for θ is

$$I(\theta) = -E_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right].$$

We have

$$\log f(x; \theta) = \log \theta + (\theta - 1) \log x,$$

so

$$\frac{\partial}{\partial \theta} \log f(x; \theta) = \frac{1}{\theta} + \log x,$$

and

$$\frac{\partial^2}{\partial \theta^2} \log f(x; \theta) = -\frac{1}{\theta^2}.$$

Therefore,

$$I(\theta) = -E \left(-\frac{1}{\theta^2} \right) = \frac{1}{\theta^2}.$$

For an unbiased estimator of $1/\theta$, the Cramér–Rao lower bound is

$$\frac{\left(\frac{\partial}{\partial \theta} (1/\theta) \right)^2}{nI(\theta)} = \frac{(-1/\theta^2)^2}{n/\theta^2} = \frac{1/\theta^4}{n/\theta^2} = \frac{1}{n\theta^2}.$$

(d) The MLE of $1/\theta$ is $-\frac{1}{n} \sum_{i=1}^n \log X_i$, which is unbiased as shown above. The variance of the MLE is

$$\text{Var} \left(-\frac{1}{n} \sum_{i=1}^n \log X_i \right) = \frac{1}{n} \text{Var}(-\log X_1).$$

Since $-\log X_1 \sim \text{Exp}(\theta)$, its variance is $1/\theta^2$. Thus,

$$\text{Var} \left(-\frac{1}{n} \sum_{i=1}^n \log X_i \right) = \frac{1}{n\theta^2},$$

which equals the Cramér–Rao lower bound. Therefore, the MLE is the best unbiased estimator for $1/\theta$.

5. Note that $X_i - \theta$ follows the $\text{Exp}(1)$ distribution, and some of the following derivations can be simplified by using this property. Nevertheless, this solution is written without using this property.

- (a) • For $\hat{\theta}_1 = X_1 - 1$:

The expectation of X_1 is

$$\begin{aligned} E(X_1) &= \int_{\theta}^{\infty} x e^{-(x-\theta)} dx \\ &= \int_0^{\infty} (y + \theta) e^{-y} dy \quad (y = x - \theta) \\ &= \int_0^{\infty} y e^{-y} dy + \theta \int_0^{\infty} e^{-y} dy \\ &= 1 + \theta. \end{aligned}$$

Therefore,

$$E(\hat{\theta}_1) = E(X_1) - 1 = (\theta + 1) - 1 = \theta.$$

- For $\hat{\theta}_2 = \bar{X} - 1$:

By linearity of expectation,

$$E(\hat{\theta}_2) = E(\bar{X}) - 1 = E(X_1) - 1 = \theta.$$

- For $\hat{\theta}_3 = \min\{X_1, \dots, X_n\} - \frac{1}{n}$:

Let $M = \min\{X_1, \dots, X_n\}$. The cumulative distribution function (CDF) of X_i is

$$F_X(x) = \int_{\theta}^x e^{-(t-\theta)} dt = 1 - e^{-(x-\theta)}, \quad x > \theta.$$

Therefore, the CDF of M is

$$\begin{aligned} F_M(m) &= P(M \leq m) = 1 - P(X_1 > m, \dots, X_n > m) \\ &= 1 - \prod_{i=1}^n P(X_i > m) \\ &= 1 - (P(X_1 > m))^n \\ &= 1 - (e^{-(m-\theta)})^n = 1 - e^{-n(m-\theta)}, \quad m > \theta. \end{aligned}$$

The density is

$$f_M(m) = \frac{d}{dm} F_M(m) = n e^{-n(m-\theta)}, \quad m > \theta.$$

The expectation of M is

$$\begin{aligned} E(M) &= \int_{\theta}^{\infty} m \cdot n e^{-n(m-\theta)} dm \\ &= \int_0^{\infty} (y + \theta) n e^{-ny} dy \quad (y = m - \theta) \\ &= n \int_0^{\infty} y e^{-ny} dy + n\theta \int_0^{\infty} e^{-ny} dy \\ &= n \cdot \frac{1}{n^2} + n\theta \cdot \frac{1}{n} \\ &= \frac{1}{n} + \theta. \end{aligned}$$

Therefore,

$$E(\hat{\theta}_3) = E(M) - \frac{1}{n} = \theta + \frac{1}{n} - \frac{1}{n} = \theta.$$

Thus, all three estimators are unbiased for θ .

- (b) • For $\hat{\theta}_1 = X_1 - 1$:

The variance of X_1 is

$$\begin{aligned} \text{Var}(X_1) &= E(X_1^2) - (E(X_1))^2 \\ E(X_1^2) &= \int_{\theta}^{\infty} x^2 e^{-(x-\theta)} dx \\ &= \int_0^{\infty} (y + \theta)^2 e^{-y} dy \quad (y = x - \theta) \\ &= \int_0^{\infty} (y^2 + 2y\theta + \theta^2) e^{-y} dy \\ &= \int_0^{\infty} y^2 e^{-y} dy + 2\theta \int_0^{\infty} y e^{-y} dy + \theta^2 \int_0^{\infty} e^{-y} dy \\ &= 2 + 2\theta \cdot 1 + \theta^2 \cdot 1 \\ &= \theta^2 + 2\theta + 2. \end{aligned}$$

Therefore,

$$\text{Var}(X_1) = (\theta^2 + 2\theta + 2) - (\theta + 1)^2 = (\theta^2 + 2\theta + 2) - (\theta^2 + 2\theta + 1) = 1.$$

So,

$$\text{Var}(\hat{\theta}_1) = \text{Var}(X_1) = 1.$$

- For $\hat{\theta}_2 = \bar{X} - 1$:

Since the X_i are independent and identically distributed,

$$\text{Var}(\bar{X}) = \frac{1}{n} \text{Var}(X_1) = \frac{1}{n}.$$

Thus,

$$\text{Var}(\hat{\theta}_2) = \text{Var}(\bar{X}) = \frac{1}{n}.$$

- For $\hat{\theta}_3 = M - \frac{1}{n}$:

Recall M has density $f_M(m) = ne^{-n(m-\theta)}$ for $m > \theta$.

The second moment is

$$\begin{aligned} E(M^2) &= \int_{\theta}^{\infty} m^2 ne^{-n(m-\theta)} dm \\ &= \int_0^{\infty} (y + \theta)^2 ne^{-ny} dy \quad (y = m - \theta) \\ &= n \int_0^{\infty} (y^2 + 2y\theta + \theta^2) e^{-ny} dy \\ &= n \left[\int_0^{\infty} y^2 e^{-ny} dy + 2\theta \int_0^{\infty} y e^{-ny} dy + \theta^2 \int_0^{\infty} e^{-ny} dy \right] \end{aligned}$$

We use the following integrals:

$$\int_0^\infty y^2 e^{-ny} dy = \frac{2}{n^3}, \quad \int_0^\infty y e^{-ny} dy = \frac{1}{n^2}, \quad \int_0^\infty e^{-ny} dy = \frac{1}{n}.$$

Therefore,

$$\begin{aligned} E(M^2) &= n \left(\frac{2}{n^3} + 2\theta \cdot \frac{1}{n^2} + \theta^2 \cdot \frac{1}{n} \right) \\ &= \frac{2}{n^2} + \frac{2\theta}{n} + \theta^2. \end{aligned}$$

Recall $E(M) = \theta + \frac{1}{n}$, so

$$\text{Var}(M) = E(M^2) - (E(M))^2 = \left(\frac{2}{n^2} + \frac{2\theta}{n} + \theta^2 \right) - \left(\theta^2 + \frac{2\theta}{n} + \frac{1}{n^2} \right) = \frac{1}{n^2}.$$

Therefore,

$$\text{Var}(\hat{\theta}_3) = \text{Var}(M) = \frac{1}{n^2}.$$